

**Operator browsers**  
Vue 3 SPA · REST (fetch) + Socket.IO (WebSocket, long-polling fallback)

HTTP + WebSocket

**Ingress** (ingress-nginx, Minikube addon) · host board.local  
/ → web:80 /api, /socket.io → api:3000 sticky sessions: cookie route (needed by Socket.IO long-polling)

### Deployment web

**web pod**

Nginx, static bundle

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Nginx, static bundle

2 replicas, PDB minAvailable: 1  
probes GET /  
preStop: sleep 5

**Secret** jwt-secret

shared by every api pod

**metrics-server**

addon, CPU metrics for HPA

**Deployment api** · 3–10 replicas (HPA, CPU 80 %), stateless w.r.t. domain state

**api pod 1**

Express — REST /api/\*

Socket.IO + Redis adapter

Change Stream listener

Leader election / SLA task

**api pod 2**

Express — REST /api/\*

Socket.IO + Redis adapter

Change Stream listener

Leader election / SLA task

**api pod N**

Express — REST /api/\*

Socket.IO + Redis adapter

Change Stream listener

Leader election / SLA task

liveness /api/health (shallow) · readiness /api/ready (Mongo + Redis + Change Stream)  
PDB maxUnavailable: 1 · JWT\_SECRET injected from the Secret

SET NX PX  
EVAL (Lua)

cross-pod  
broadcast

reads / writes  
(to the primary)

Change Stream  
(one per replica)

**Job** mongo-rs-init  
idempotent rs.initiate()

### Redis (1 replica, no persistence)

**Leadership**

leader:escalation  
TTL 15s

**Claim locks**

lock:incident:<id>  
TTL 60s

**Pub/Sub**

Socket.IO  
adapter

single point of failure (deliberate trade-off)  
check-and-act made atomic by Lua EVAL

### MongoDB replica set rs0

**mongo-0**

PVC 1 Gi

**mongo-1**

PVC 1 Gi

**mongo-2**

PVC 1 Gi

StatefulSet + headless Service mongo · PDB minAvailable: 2  
collections incidents, users · primary elected automatically

→ request / command    - - - → event / notification    → configuration / operations

- - - - - Kubernetes cluster (single-node Minikube), namespace app